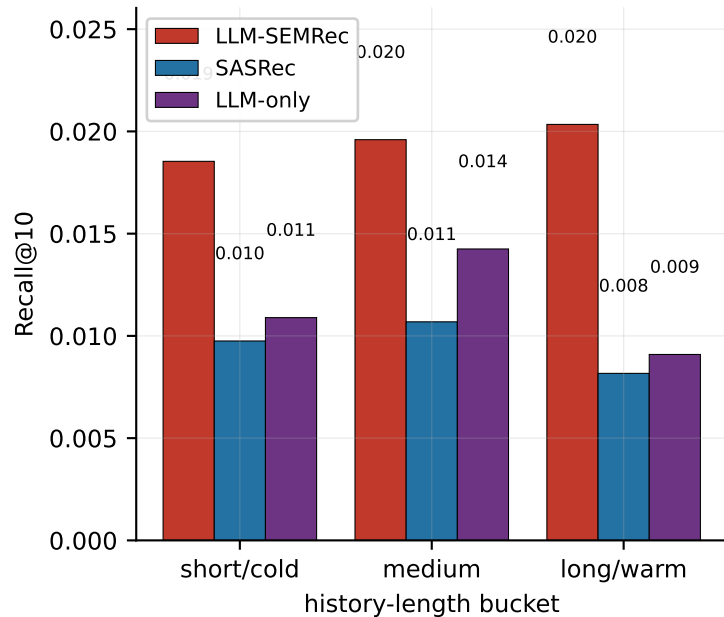
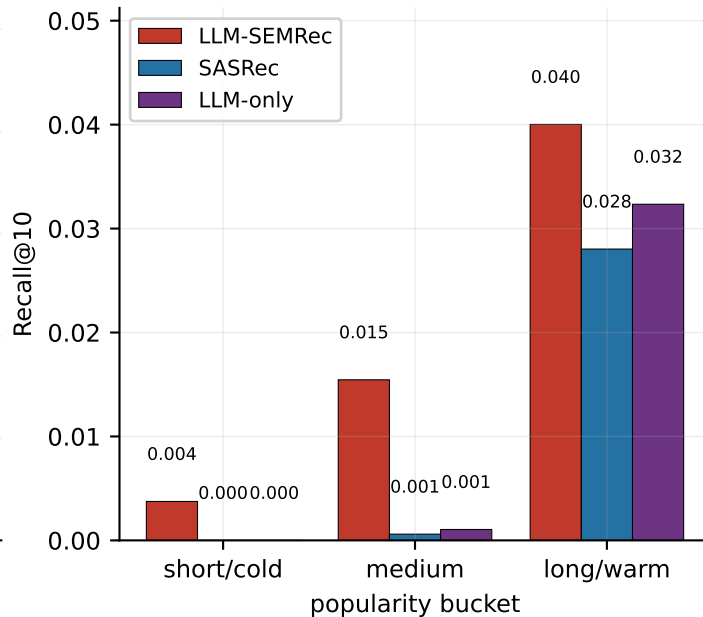


Sparsity and cold-start analyses (Amazon Beauty)

(a) Sparse feedback: by history length



(b) Cold start: target-item popularity



(c) Long-tail items

